

Israt Zarin

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104D Avery Hall, University of Nebraska- Lincoln, Lincoln, NE, USA

EDUCATION

Spring 24 – Present University of Nebraska-Lincoln, Ph.D.
Major: Computer Science
Expected: 2028

August 2026 University of Nebraska-Lincoln,
Master of Science (M.Sc.) in
Computer Science

May 2021 Jahangirnagar University (JU),
Master of Science (M.Sc.) in
Computer Science

May 2019 American International
University-Bangladesh (AIUB),
Bachelor of Computer Science and
Engineering

ACADEMIC EXPERIENCE

Summer 2025-Present Graduate Research Assistant
School of Natural Resource, University of Nebraska-Lincoln

Spring 2024-Spring 2025 Graduate Teaching Assistants
Computer Science, University of Nebraska-Lincoln

2020-2023 Lecturer
Computer Science, Uttara University, Dhaka, Bangladesh

SKILL

Hard Skill:

- GIS Tool: Arc GIS
- Web Technology: Laravel, Nodejs, Html5, CSS, JavaScript, PHP, XML, json, jQuery, Ajax, WordPress
- Programming Language: C, C++, C#, Java, Python
- GIS Tool: Arc GIS
- Database: SQL Server, MySQL, Oracle
- Machine learning framework: TensorFlow, Pytorch, Keras
- Engineering Drawing Tool: AutoCAD
- MS tool: MS word, excel, power point, access

Soft Skill:

- Communication skills
- Problem solving
- Teamwork/collaboration

RESEARCH INTEREST

- GIS Data Analysis
- Machine Learning
- Image Processing
- Natural Language Processing
- Wireless Network security
- Cryptography
- Education
- Image Processing

PUBLICATIONS

- "SWINC: Secured Wireless Intelligent Network Using Cryptography "ICO2022(Published) [Link!](#)
- "Execution examination of different edge detection algorithms" (IEEE)ISMODE 2022" (Published) [Link!](#)
- "Evaluate Performance of Several Edge Detection Techniques on Medical Images" (Published)[Link!](#)

CURRENT PROJECT

Developed a context-aware meta-learning framework using GIS-based Nebraska lake visitation data for the Nebraska Game and Parks Commission to improve visitor forecasting. Different models usually succeed for different reasons, context, under different conditions. No single model can perform uniformly in all contexts, for example, there must be differences in terms of spatial (static) and temporal context (Dynamic). Motivated by this observation, our research work is beyond this one model fit strategies. This research project investigates why different predictive models succeed under different conditions, for instance different behavioral and environmental context, rather than assuming a single model works optimally in any condition. For example, Tree based models like Random Forest, XGBoost can capture spatial heterogeneity such as demographics of visitor and static attributes but underperforms under dynamic attribute such as visitation in term of weekend, weekdays, seasons and so on. To address this, we proposed a context aware meta learning framework that can switch to the model in terms of interpretable features such as season, weekend indicators, demographic dominance (e.g., high-income or senior-heavy lakes), and intra-day visitation variability. This framework integrates rule-guided reasoning with a lightweight Random Forest meta-classifier that decides which model should be selected after analyzing type of features. For example, if we provide any interpretable features like 'visitor forecasting for first Friday of summer in Homes Lake', it will switch to the extract meta model after analyzing the interpretable feature like 'Summer', 'Weekend' and so on. Real-world data, such as visitation to the lake, does not always act uniformly or places. There are busy lakes, quiet lakes and the behavior of people vary depending on seasons, weekends and local conditions. In many cases, one can make inaccurate and unrealistic predictions based on a single model that is used in all cases. This research is important because it presents a simple and intuitive concept: apply various models in various situations, depending on clear and intelligible context like season, type of lakes, and visitor trends. This method is particularly applicable to such fields as natural resource and fisheries management, environmental planning, tourism and public policy where a decision will impact on an ecosystem and pressure management such as weekend pressure, holiday pressure and predictions will have to be not only correct but also simple to comprehend and trust. For the experiment, GIS data is used.

REFERENCE

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